ललितपुर उपमहानगरपालिका कार्यालय पदपूर्ति समिति
इन्जिनियरिङ सेवा, अधिकृतस्तर छैटौ तह स्त्रस्चरल इन्जिनियरिङ पद
खुला प्रतियोगितात्मक लिखित परीक्षा
मिति: 2072/11/29

समय: 3 घण्टा पूर्णाङ्क: 100

1. **Differentiate between stiffness matric and flexibility matrix methods.**

Answer:

Both are matrix methods of structural analysis, but they differ mainly in the basic unknowns used in the analysis.

Stiffness Matrix Method	Flexibility Matrix Method
The basic unknowns in stiffness matrix method are Joint displacements (translations and rotations)	The basic unknowns in flexibility matrix method are redundant forces.
The primary relationship of stiffness matrix method is {P}=[K]{D}	The primary relationship of flexibility matric method is represented by {D}=[F]{P}
The matrix used is Stiffness matrix [K]	The matrix used is Flexibility matrix [F]
In stiffness matrix method, applied loads are generally known.	In flexibility matrix method, applied loads and primary structure responses are known.
The unknown quantities to be determined are nodal displacements and then member forces	The unknown quantities to be determined are redundant forces and then member forces
Stiffness matrix method is best suited for indeterminate and large structures.	Flexibility matrix method is best suited for the structures with few redundant.
Satisfied through displacement formulation	Compatibility equations are explicitly used
Very widely used; forms the basis of the Finite Element Method (FEM)	Less convenient for large structures
Depends mainly on number of degrees of freedom	Depends mainly on number of redundant

2. **What is concrete admixture? Briefly explain about use of prestress concrete.**

Answer:

A concrete admixture is a material other than cement, water, and aggregates that is added to concrete before or during mixing in a small quantity to modify one or more properties of fresh or hardened concrete.

Common types and uses:

- **Accelerating admixtures:** Reduce setting time and increase early strength.
- **Retarding admixtures:** Delay setting, useful in hot weather and long-distance transportation.
- **Water-reducing admixtures (plasticizers):** Reduce water requirement and improve workability.
- **Superplasticizers:** Produce highly workable concrete with a low water-cement ratio.
- **Air-entraining admixtures:** Introduce tiny air bubbles to improve resistance to freezing and thawing.
- **Waterproofing admixtures:** Reduce permeability and improve water resistance.

Prestressed concrete is the type of concrete in which internal compressive stresses are introduced by tensioning high-strength steel tendons before or after loading, so that tensile stresses caused by external loads are reduced or eliminated.

Prestressed concrete is widely used for bridges, flyovers, long-span roofs, tanks, and other structures requiring high strength, durability, and longer spans, despite its higher initial cost and construction complexity.

Major uses of prestressed concrete are:

- **Bridges and flyovers**: for long-span beams and girders.
- **Building floors and roofs**: to achieve longer spans with thinner sections.

- **Railway sleepers**: to resist repeated and dynamic loads.
- **Water tanks and pressure pipes**: to minimize cracking and leakage.
- **Industrial buildings and auditoriums**: for large column-free spaces.
- **Poles and precast structural members**: where high strength and durability are required.

3. **What are the main components of analysis of item rates? Give an example of analysis of an item rate. [10]**

Answer:

Rate analysis is the process of determining the unit cost of a particular construction item by considering the quantities and costs of materials, labor, equipment, overheads and profit required to complete one unit of work.

The main components are:

i. **Material cost**: Cost of cement, sand, aggregate, bricks, steel, etc., including transportation and wastage where applicable.
ii. **Labor cost**: Wages of skilled, semi-skilled and unskilled workers required for the work.
iii. **Equipment/Tools and plants**: Cost of machinery, equipment, scaffolding, tools, etc., including their operation and maintenance.
iv. **Transportation/Carriage**: Cost of transporting materials from the source to the construction site.
v. **Overhead charges**: Site and establishment expenses, supervision, water, electricity, office expenses, etc.
vi. **Contractor's profit**: Reasonable profit added to the cost of execution.
vii. **Taxes and other applicable charges**: Government taxes, royalties, fees, etc., as applicable.

For **1 m^3 of cement concrete 1:2:4**, approximately **1.54 m^3 of dry materials** are required.

Particular	Approx. quantity	Rate	Amount
Cement	0.22 m^3 (≈ 6.34 bags)	Rs. 800/bag	Rs. 5,072
Sand	0.44 m^3	Rs. 2,500/m^3	Rs. 1,100
Coarse aggregate	0.88 m^3	Rs. 2,200/m^3	Rs. 1,936
Labour	—	—	Rs. 2,000
Tools & equipment	—	—	Rs. 300
Subtotal			**Rs. 10,408**
Overhead & profit (15%)			Rs. 1,561
Total rate			**≈ Rs. 11,969/m^3**

4. **What is Quality Assurance Plan? Explain it with example? [10]**

Answer:

A Quality Assurance Plan (QAP) is a systematic document that defines the procedures, responsibilities, standards, inspections, tests, and controls required to ensure that construction work meets the specified quality, performance, and safety requirements.

It is prepared before or during project execution to ensure that quality is planned and controlled throughout the project, rather than checked only after the work is completed.

Main Contents of a QAP:

i. **Quality objectives –** Standards and quality requirements to be achieved.
ii. **Applicable codes and specifications –** Relevant standards such as IS, ASTM, BS, NBC, etc.
iii. **Material control –** Inspection and testing of cement, aggregates, steel, bricks, etc.
iv. **Work procedures –** Approved methods for carrying out construction activities.
v. **Inspection and testing plan (ITP) –** Tests, frequency, acceptance criteria and responsible personnel.

vi. **Responsibilities –** Duties of contractor, consultant, quality-control personnel and other parties.
vii. **Documentation and records –** Test reports, inspection records, material approvals and non-conformance reports.
viii. **Corrective actions –** Procedures for identifying and correcting defective or non-conforming work.

Example: QAP for Reinforced Cement Concrete (RCC) Work

For an RCC beam, the QAP may include:

Activity	Quality check/test
Cement and aggregates	Material testing
Reinforcement steel	Check diameter, grade and spacing
Formwork	Check dimensions, alignment and supports
Concrete mixing	Check mix proportion and workability
Slump test	Slump test before placing concrete
Concrete strength	Cube compressive-strength test
Curing	Check curing duration and method
Final inspection	Check dimensions, cracks and defects

5. **A. What are the provisions relating to the construction of building as per Local Government Act 2055 and regulation 2056?**

Answer:

Under the Local Self-Governance Act, 2055 (1999) and the Local Self-Governance Rules, 2056, building construction within a municipality or Village Development Committee area was regulated through prior approval, approved drawings, inspection, and enforcement. The main provisions were contained in Chapter 9, Sections 149–163 of the Act.

i. Prior approval is compulsory

No person may construct a building within a municipal area without obtaining construction approval from the municipality.

Approval is required for:

- Constructing a new building.
- Reconstructing a demolished or damaged building.
- Changing the front portion or external appearance of a building.
- Altering an approved building design or layout.

ii. Application for construction approval

A person wishing to construct a building must submit an application to the concerned local body in the prescribed form. The application should generally include:

- Building drawings and architectural plans.
- Structural design and other technical details.
- Site plan, location plan, floor plans, elevations, and sections.
- Details of the building area, number of floors, length, and breadth.
- Consent of the landowner where the applicant is not the owner.

iii. Approval of drawings and design

The municipality must approve the building drawings and design before construction begins. The design must comply with:

- Applicable building bye-laws.
- Land-use and settlement-development plans.
- Road and street widths.
- Public health and sanitation requirements.
- Structural safety requirements.
- National or locally applicable building standards.

The municipality was also empowered to approve or arrange for the approval of designs of houses and buildings constructed within the municipal area.

iv. Construction according to the approved plan

The owner must construct the building strictly according to the approved drawings and design. Without obtaining additional approval, the owner must not:

- Add another floor.
- Increase the length or breadth of the building.
- Change the approved façade or front elevation.
- Change the use or layout of the building.
- Construct additional structures not shown in the approved plan.

A change that affects the size, height, floor area, or external appearance generally requires revised approval.

v. Building construction period

Construction was required to be completed within the period prescribed by the Act and Rules. The earlier provision generally allowed completion within two years from the date of approval.

If construction could not be completed within that period, the owner had to apply for an extension. An extension could be granted for a further prescribed period, subject to payment of the applicable additional fee or penalty.

vi. Inspection by the local authority

The municipality or Village Development Committee could inspect the building:

- Before construction.
- During foundation and structural construction.
- During other important stages.
- After completion.

Inspection was intended to verify whether:

- Construction approval had been obtained.
- The work followed the approved drawings.
- The building occupied only the approved plot area.
- Public land, roads, drains, or rights-of-way had not been encroached upon.
- The building complied with safety and construction requirements.

vii. Action against unauthorized construction

If a person constructed a building without approval or contrary to the approved plan, the local authority could:

- Order the construction to stop.
- Direct the owner to correct the deviation.
- Impose a fine or additional charge.
- Order demolition of the unauthorized portion.

- Demolish the building or unauthorized part itself.

6. **In recent Gorkha earthquake, significant number of temples and monumental structures were severely damaged in Kathmandu Valley. As a structural engineer of LSMC, what would be your recommendation to improve the seismic strength of such temples and structures to prevent damages in future earthquake? Shall modern construction materials permitted to be used in repair and restoration of such structures.**

Answer:

The 2015 Gorkha Earthquake caused extensive damage to many historic temples and monuments in the Kathmandu Valley. Therefore, as a structural engineer of Lalitpur Metropolitan City (LSMC), I would recommend a conservation-based seismic strengthening approach, where structural safety is improved without compromising the historical, architectural and cultural value of the monument. UNESCO has also emphasized that recovery should be based on proper documentation, analysis of damage, and appropriate traditional materials and techniques.

i. Detailed assessment and documentation

- Conduct a structural and architectural survey of each monument.
- Record cracks, tilting, settlement, deformation and damaged structural members.
- Prepare detailed drawings, photographs and condition maps before intervention.
- Carry out material testing and investigation of foundations, masonry and timber.
- Identify original and later-added elements before deciding on strengthening.

ii. Improve the structural integrity

- Strengthen weak brick masonry walls and repair cracks using compatible traditional mortar.
- Improve the connection between walls, floors, roofs and timber members so that the structure acts as an integrated unit during an earthquake.
- Provide discreet timber ties, bands and diaphragms where historically appropriate.
- Strengthen deteriorated timber members by splicing, sistering or other minimally invasive techniques.

iii. Reduce earthquake vulnerability

- Remove unnecessary additional loads and poorly executed previous alterations.
- Improve the load path from roof to foundation.
- Provide adequate lateral resistance against earthquake forces.
- Where necessary, use concealed reinforcement or seismic devices only after structural analysis and approval by the concerned heritage authorities.

iv. Use traditional construction techniques

- Traditional brick, timber and lime-surkhi mortar should generally be preferred because they are compatible with the existing structure and help preserve authenticity.
- UNESCO's restoration of the Radha Krishna Temple in Kathmandu, for example, prioritized traditional materials, traditional sealant and lime-surkhi mortar, while carefully documenting and reusing original elements.

v. Proper maintenance and monitoring

- Carry out periodic inspection of cracks, timber deterioration, moisture and foundation settlement.
- Control water leakage and vegetation growth.
- Establish a regular maintenance programme.
- Install monitoring systems where necessary to detect progressive deformation.

UNESCO has specifically identified poor maintenance, inadequate documentation and inappropriate restoration practices as important issues affecting heritage structures.

Yes, but only selectively and under strict conservation principles. Modern materials should not automatically replace traditional materials**.** Their use should be permitted only when:

- The traditional method cannot provide the required structural safety.
- The intervention is necessary, compatible, reversible or minimally invasive as far as practicable.
- It does not adversely affect the monument's historical appearance or authenticity.
- It has been justified through structural analysis and approved by the Department of Archaeology and relevant heritage authorities.
- The modern material is compatible with the existing brick, timber and mortar and will not cause long-term deterioration.

Kasthamandap, the historic wooden pavilion that gave Kathmandu its name, collapsed completely during the 2015 earthquake. The collapsed structure was carefully documented and surviving historical materials were salvaged. Traditional construction methods and craftsmanship were used in reconstruction. Reconstruction was undertaken with attention to seismic safety.

Another example is Dharahara. The historic tower collapsed during the 2015 earthquake, killing many people. The tower experienced earthquake-related collapse before: the original tower was built in 1832 and was reconstructed after the 1833 earthquake. It was again severely damaged/collapsed in 2015. The new Dharahara was constructed with a modern structural system, while retaining the recognizable historic appearance.

7. **For a saturated soil, given w= 40% and G=2.71, determine the saturated and dry unit weight of soil.**

Answer:

Given,

$$w = 40\% = 0.40$$
$$G = 2.71$$
$$\gamma_w = 9.81\,\text{kN/m}^3$$

$$\textit{Since the soil is saturated},$$
$$S = 1$$

$$\textit{For saturated soil},$$
$$w = \frac{Se}{G}$$

Therefore,

$$e = wG$$
$$e = 0.40 \times 2.71 = 1.084$$

The saturated unit weight is given by:

$$\gamma_{\text{sat}} = \frac{G+e}{1+e}\gamma_w$$

$$\textit{Substituting the values},$$
$$\gamma_{\text{sat}} = \frac{2.71+1.084}{1+1.084} \times 9.81$$
$$\gamma_{\text{sat}} = \frac{3.794}{2.084} \times 9.81$$
$$\gamma_{\text{sat}} \approx 17.85\,\text{kN/m}^3$$

The dry unit weight is given by:

$$\gamma_d = \frac{G}{1+e}\gamma_w$$

Substituting the values,

$$\gamma_d = \frac{2.71}{1+1.084} \times 9.81$$
$$\gamma_d = \frac{2.71}{2.084} \times 9.81$$
$$\gamma_d \approx 12.76\ \mathrm{kN/m^3}$$
$$\gamma_{\mathrm{sat}} = 17.85\ \mathrm{kN/m^3},$$

8. **A 5 m high retaining wall having angle of repose 30-degree, C= 5 KN/m^2 and unit weight 17 KN/m^3. Determine the active pressure on the wall a. before the formation of crack b. After the formation of crack.**
Answer:
Given,

$$H = 5m$$
$$\phi = 30^\circ$$
$$c = 5\,kN/m^2$$
$$\gamma = 17\,kN/m^3$$

$$\text{The active earth pressure coefficient is}$$
$$K_a = \frac{1-\sin\phi}{1+\sin\phi}$$
$$K_a = \frac{1-\sin 30^\circ}{1+\sin 30^\circ}$$
$$K_a = \frac{1-0.5}{1+0.5}$$
$$= \frac{1}{3}$$
$$= 0.333$$
$$\sqrt{K_a} = \sqrt{\frac{1}{3}} = 0.577$$
$$(a)\ \text{Active pressure before formation of crack}$$

$$\text{For cohesive soil, active lateral pressure at depth } (z) \text{ is}$$

$$\sigma_a = K_a \gamma z - 2c\sqrt{K_a}$$

$$\text{At the bottom of the retaining wall, } (z = H = 5m),$$

$$\sigma_a = \frac{1}{3}(17)(5) - 2(5)\sqrt{\frac{1}{3}}$$
$$\sigma_a = 28.333 - 5.774$$
$$\sigma_a = 22.56\,kN/m^2$$

$$\text{At the top of the wall,}$$

$$\sigma_{a,\mathrm{top}} = -2c\sqrt{K_a}$$
$$\sigma_{a,\mathrm{top}} = -2(5)(0.577) = -5.77\,kN/m^2$$

$$Therefore, \; the \; total \; active \; pressure \; is$$

$$P_a = \frac{1}{2}(-5.77 + 22.56)(5)$$
$$P_a \approx 41.97 \, kN/m$$
$$(b) \; Active \; pressure \; after \; formation \; of \; crack$$

$$The \; depth \; of \; tension \; crack \; is \; given \; by$$
$$z_c = \frac{2c}{\gamma \sqrt{K_a}}$$

$$Substituting \; the \; values,$$
$$z_c = 1.019 \, m$$

$$The \; effective \; height \; of \; soil \; producing \; active \; pressure \; is$$

$$H' = H - z_c$$
$$H' = 5 - 1.019$$
$$H' = 3.981 \, m$$

$$The \; active \; pressure \; at \; the \; bottom \; after \; crack \; formation \; is$$

$$\sigma_a = K_a \gamma H' - 2c\sqrt{K_a}$$
$$\sigma_a = \frac{1}{3}(17)(3.981) - 2(5)(0.577)$$
$$\sigma_a = 22.559 - 5.774$$
$$\sigma_a = 16.80 \, kN/m^2$$

$$Therefore, \; the \; total \; active \; pressure \; is$$

$$P_a = \frac{1}{2}\sigma_a H'$$
$$P_a = \frac{1}{2}(16.80)(3.981)$$
$$P_a \approx 33.44 \, kN/m$$

9. **Determine the ultimate moment capacity. By Limit State Method, of a rectangular concrete beam section of 300 mm width and 550 mm effective depth, with 3 numbers of 20 mm diameter Fe steel as tension reinforcement. Take M20 as the grade of the concrete and 25 mm clear cover.**

   Answer:

   Given,

$$b = 300 \, mm$$
$$d = 550 \, mm$$
$$f_{ck} = 20 \, N/mm^2$$
$$f_y = 415 \, N/mm^2$$
$$\text{Tension reinforcement} = 3 \text{ bars of } 20 \, mm \text{ diameter}$$

$$\text{Clear cover} = 25\,mm$$

$$Step\ \ 1: Area\ \ of\ \ tension\ \ reinforcement$$

$$A_{st} = 3\left(\frac{\pi}{4} \times 20^2\right)$$

$$A_{st} = 942.48\,m\,m^2$$

$$Step\ \ 2: Determine\ \ the\ \ depth\ \ of\ \ neutral\ \ axis$$

$$For\ \ a\ \ singly\ \ reinforced\ \ rectangular\ \ section,$$

$$x_u = 0.87 f_y A_{st} / 0.36 f_{ck} b$$

$$Substituting\ \ the\ \ values,$$

$$x_u = 148.63\,mm$$

$$For\ \ Fe\ \ 415\ \ steel,$$

$$x_{u,max} = 0.48\,d$$

$$x_{u,max} = 0.48(550) = 264\,mm$$

$$Since$$

$$x_u < x_{u,max}$$

$$148.63\ \ <\ \ 264$$

Hence, the section is under-reinforced.

$$Step\,3: Ultimate\ \ moment\ \ capacity$$

$$The\ \ ultimate\ \ moment\ \ capacity\ \ is\ \ given\ \ by$$

$$M_u = 0.87 f_y A_{st}(d - 0.42\,x_u)$$

$$Substituting\ \ the\ \ values,$$

$$M_u = 0.87(415)(942.48)(550 - 0.42(148.63))$$

$$M_u = 166{,}264{,}000\,N\,mm$$

$$Converting\ \ into\ \ (kN\,m),$$

$$M_u = \frac{166{,}264{,}000}{10^6}$$

$$M_u \approx 166.26\,kN\,m$$

10. **Illustrate with sketches about the various classes of live loads used in design of vehicular bridge. What other loads in addition to live load is essential to be considered while designing superstructure of Reinforced Concrete Bridges.** **[8+2=10]**

Answer:

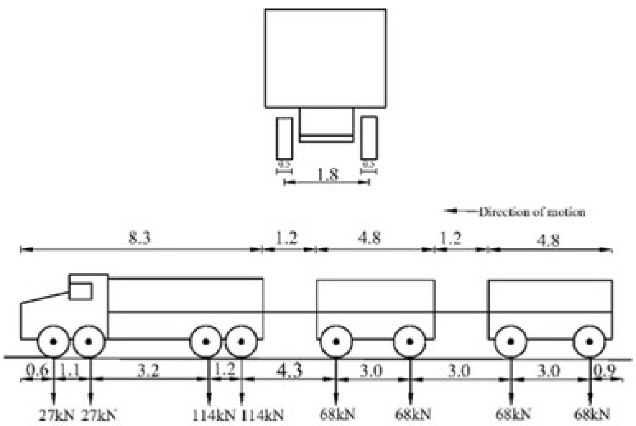
Various loads and forces acting on the bridges are discussed below:

a. Live load:

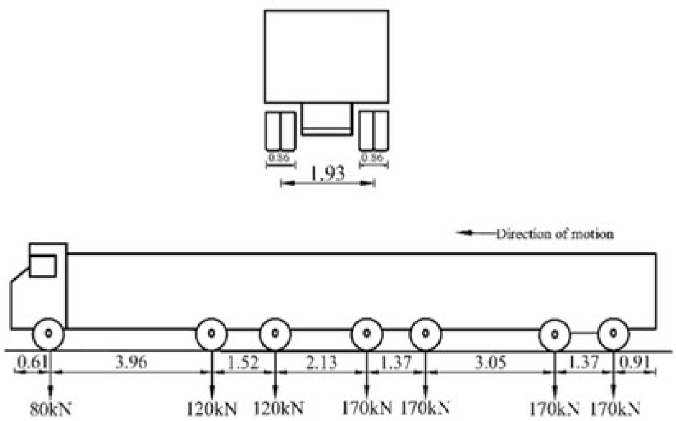
- Moving load along the length of the road is known as live load.
- According to IRC, live load can be:

i. IRC Class A loading:

- Treated as standard loading which is adopted for the design and construction of permanent bridges.



(a)



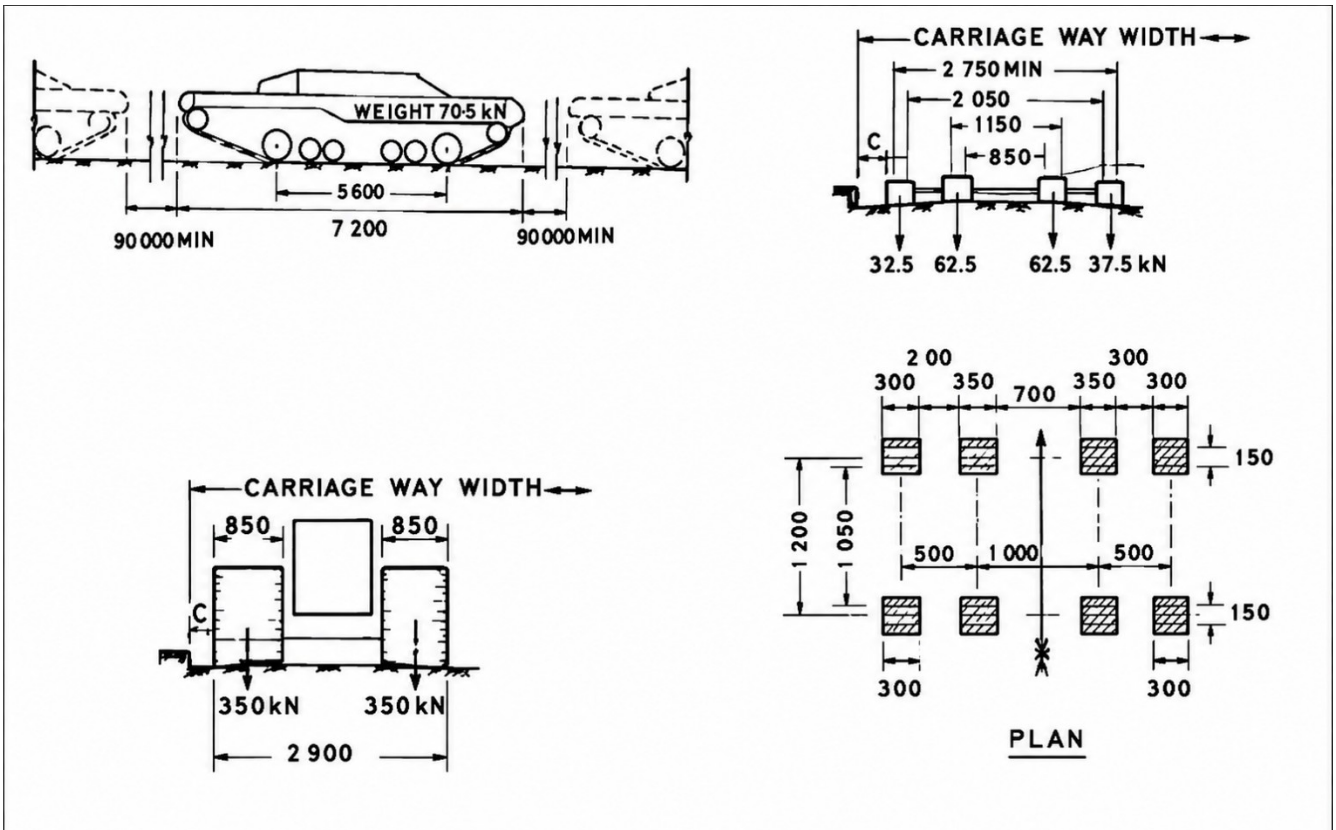
(b)

ii. IRC Class B loading:

- Adopted for the design and construction of temporary bridges and bridges in specifies areas.

iii. IRC Class AA loading:

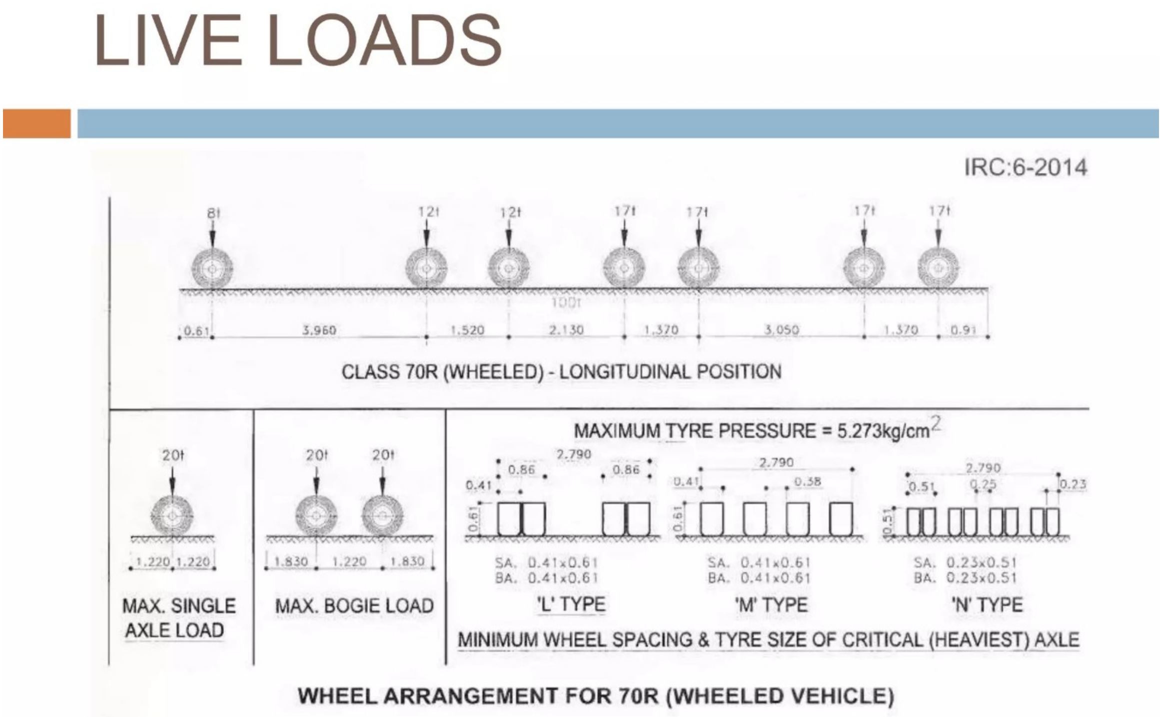
- Treated as abnormal loading used for construction of bridges where heavy traffic load is to be subjected.



(a) Tracked vehicle *(a) Wheeled vehicle*

iv. IRC Class 70R loading:

- ➢ Treated as abnormal loading, which has three patterns of loading i.e., wheeled load, tracked load and train load.



b. Impact load:

- ➢ Dynamic effect of moving live load along the length of bridge.

- Impact load is determined by using the formula:

Impact load= Impact Factor x Static value of live load

c. **Wind load:**

- Stress subjected due to wind on the exposed part of bridge.
- Wind load= Wind load on the structure + Wind load on live load.
- $F_t = P_z$ x A x G x C_D

Where F_t = Wind load in transverse directions

P_z = Design wind pressure

A= Area of the exposed part of structure

G= Gust Factor

C_D= Coefficient of Drag

d. **Longitudinal forces:**

- Horizontal forces acting parallel to the length of road.
- Longitudinal forces on the bridge occurs:
  i. Due to tracking effect through acceleration of driving wheels.
  ii. Due to braking effect through application of brakes.
  iii. Frictional resistance offered to the movement of free bearing.

e. **Lateral forces:**

- In case of highway bridge, lateral load due to parapet and kerb is normally considered.
  i. Parapet:
     Shall be designed to resist lateral horizontal forces and vertical forces each of 150 kg/m applied simultaneously on the top.
  ii. Kerbs:
     Shall be designed for lateral loadings of 750 kg/m applied horizontally on the top of kerbs.

Impact factor is the factor considered in bridge live load to account for the dynamic effects of moving vehicles, such as vibration and sudden wheel loads. The factor increases the static load so that bridge is safe under traffic conditions as:

Impact load= Impact Factor x Static value of live load

Formula of impact factor for RCC bridge:

For **RCC bridges**, the impact factor is commonly calculated using the formula:

$$IF = \frac{4.5}{6 + L}$$

Where, IF= Impact factor

L= effective span of the bridge (m)

The **impact load** is then:

$$\text{Impact Load} = IF \times \text{Live Load}$$

The **total design live load** becomes:

$$\text{Design Live Load} = \text{Live Load}\left(1 + IF\right)$$